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Foreword

This European Telecommunication Standard (ETS) has been produced by the Terminal Equipment (TE) Technical Committee of the European Telecommunications Standards Institute (ETSI).

This ETS consists of four parts as follows:

Part 1: "Coding of multimedia and hypermedia information for basic multimedia applications (MHEG-5)";

Part 2: "Use of Digital Storage Media Command and Control (DSM-CC) for basic multimedia applications";

Part 3: "Application Programmable Interface (API) for MHEG-5 ";

Part 4: "Videotex Man Machine Interface (VEMMI) enhancements to support broadband multimedia information retrieval services".

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1 Scope

This European Telecommunications Standard (ETS) specifies the MHEG part 5 Abstract Syntax Notation One (ASN.1) notation consisting of a syntax description (equivalent to the Extended Backus Naur Form (EBNF) syntax) and encoding rules.

MHEG part 5 (ISO/IEC IS 13522-5 [1]) specifies the coded representation of interchanged multimedia/hypermedia information objects (MHEG-5 objects) for use in the domain of base-level interactive applications such as movies-on-demand, teleshopping, near video-on-demand.

MHEG-5 specifies objects and their semantics using an informal text description. It also provides a formal description of the interchanged objects syntax using EBNF.

This specification is included in ISO/IEC IS 13522-5 [1] as normative annex A.

2 Normative references

This ETS incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies.

[1] ISO/IEC 13522-5 (1996): "Information technology - Coding of Multimedia and Hypermedia information - Part 5: Support for Base-Level Interactive Applications".

[2] ISO/IEC 8824-1 (1996)/ITU-T Recommendation X.680 (1995): "Information technology - Abstract Syntax Notation One (ASN.1): Specification of basic notation - Amendment 1".

[3] ISO/IEC 8825-1 (1996)/ITU-T Recommendation X.690 (1995): "Information technology - ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER)".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of this ETS, the definitions in ISO/IEC IS 13522-5 [1], ISO/IEC 8824-1 [2] and ISO/IEC 8825-1 [3] apply.

3.2 Abbreviations

For the purposes of this ETS, the following abbreviations apply:

ASN.1	Abstract Syntax Notation One
BER	Basic Encoding Rules
DER	Distinguished Encoding Rules
EBNF	Extended Backus Naur Form
MHEG	Multimedia and Hypermedia Expert Group

4 MHEG-5 ASN.1 notation design principles

Designing an ASN.1 encoding is always a trade-off between the compactness of the encoded datasyntax and the compactness of the ASN.1 parser required. As this syntax is designed to run on a minimal resource platform, the compactness of the ASN.1 parser is favoured with regard to the compactness of the interchanged code.

These constraints lead to the following design principles:

1) use of semantic constraints to allow parsers to do more conformance checking;
2) use of context-free tags to allow parsers to do strong type checking;
3) uniform structure for elementary action to allow parsers to use a general algorithm.

5 MHEG-5 ASN.1 notation

This clause defines the ASN.1 notation for the syntax of MHEG-5 objects conforming to ISO/IEC 13522-5 [1].

The encoding of the MHEG-5 objects from this ASN.1 syntax shall make use of the Distinguished Encoding Rules (DER) defined in ISO/IEC 8825-1 [3].

The syntax that shall be used in the DER is detailed below:

```
--$PREFIX=ISOMHEG-mheg-5:mheg-5
-- Module: mheg-5

ISO13522-MHEG-5
{joint-iso-itu-t(2) mheg(19) version(1) mheg-5(17)} DEFINITIONS IMPLICIT
TAGS ::=
BEGIN

-- This module defines the MHEG-5 abstract syntax which consists of data values of type:
--   ISO13522-MHEG-5.InterchangedObject.
-- This abstract syntax is identified by the name: {joint-iso-itu-t(2) mheg(19) version(1)
mheg-5(17)}.

InterchangedObject ::= CHOICE
{
  application   [0] ApplicationClass,
  scene     [1] SceneClass
}

--  Root Class ____________________________________________________

RootClass ::= ObjectReference

--  Group Class ___________________________________________________

GroupClass ::= SET
{
  RootClass (WITH COMPONENTS
    {external-reference (WITH COMPONENTS {..., object-number (0)}) PRESENT,
     internal-reference ABSENT}),
  standard-identifier   [2] StandardIdentifier OPTIONAL,
  standard-version  [3] INTEGER (1) OPTIONAL,
  object-information    [4] OCTET STRING OPTIONAL,
  on-start-up   [5] ActionClass OPTIONAL,
  on-close-down     [6] ActionClass OPTIONAL,
  original-group-cache-priority     [7] INTEGER (0..255) DEFAULT 127,
  items     [8] SEQUENCE SIZE (1..MAX) OF GroupItem OPTIONAL
}

StandardIdentifier ::= SEQUENCE
{
  joint-iso-itu     INTEGER (2),
  mheg  INTEGER (19)
}

GroupItem ::= CHOICE
{
  resident-program  [9] ResidentProgramClass,
  remote-program    [10] RemoteProgramClass,
  interchanged-program  [11] InterchangedProgramClass,
  palette   [12] PaletteClass,
  font  [13] FontClass,
  cursor-shape  [14] CursorShapeClass,
```

```
  boolean-variable  [15] BooleanVariableClass,
  integer-variable  [16] IntegerVariableClass,
  octet-string-variable     [17] OctetStringVariableClass,
  object-ref-variable   [18] ObjectRefVariableClass,
  content-ref-variable  [19] ContentRefVariableClass,
  link  [20] LinkClass,
  stream    [21] StreamClass,
  bitmap    [22] BitmapClass,
  line-art  [23] LineArtClass,
  dynamic-line-art  [24] DynamicLineArtClass,
  rectangle     [25] RectangleClass,
  hotspot   [26] HotspotClass,
  switch-button     [27] SwitchButtonClass,
  push-button   [28] PushButtonClass,
  text  [29] TextClass,
  entry-field   [30] EntryFieldClass,
  hyper-text    [31] HyperTextClass,
  slider    [32] SliderClass,
  token-group   [33] TokenGroupClass,
  list-group    [34] ListGroupClass
}

--. Application Class ____________________________________________

ApplicationClass ::= SET
{
  COMPONENTS OF     GroupClass,
  on-spawn-close-down   [35] ActionClass OPTIONAL,
  on-restart    [36] ActionClass OPTIONAL,
  default-attributes    [37] SEQUENCE SIZE (1..MAX) OF DefaultAttribute OPTIONAL
}

DefaultAttribute ::= CHOICE
{
  character-set     [38] INTEGER,
  background-colour     [39] Colour,
  text-content-hook     [40] INTEGER,
  text-colour   [41] Colour,
  font  [42] FontBody,
  font-attributes   [43] OCTET STRING,
  interchanged-program-content-hook     [44] INTEGER,
  stream-content-hook   [45] INTEGER,
  bitmap-content-hook   [46] INTEGER,
  line-art-content-hook     [47] INTEGER,
  button-ref-colour     [48] Colour,
  highlight-ref-colour  [49] Colour,
  slider-ref-colour     [50] Colour
}

FontBody ::= CHOICE
{
  direct-font   OCTET STRING,
  indirect-font     ObjectReference
}

--  Scene Class _________________________________________________

SceneClass ::= SET
{
  COMPONENTS OF GroupClass,
  input-event-register  [51] INTEGER,
  scene-coordinate-system   [52] SceneCoordinateSystem,
  aspect-ratio  [53] AspectRatio DEFAULT {width 4, height 3},
  moving-cursor     [54] BOOLEAN DEFAULT FALSE,
  next-scenes   [55] SEQUENCE SIZE (1..MAX) OF NextScene OPTIONAL
}

SceneCoordinateSystem ::= SEQUENCE
{
  x-scene   INTEGER,
  y-scene   INTEGER
}

AspectRatio ::= SEQUENCE
{
  width  INTEGER,
  height    INTEGER
}

NextScene ::= SEQUENCE
{
  scene-ref     OCTET STRING,
  scene-weight  INTEGER (0..255)
}
-- . Ingredient Class _____________________________________________
```

```
 IngredientClass ::= SET
{
  RootClass (WITH COMPONENTS
    {..., external-reference (WITH COMPONENTS {..., object-number (1..MAX)})}),
  initially-active  [56] BOOLEAN DEFAULT TRUE,
  content-hook  [57] INTEGER OPTIONAL,
  original-content  [58] ContentBody OPTIONAL,
  shared    [59] BOOLEAN DEFAULT FALSE
}

ContentBody ::= CHOICE
{
  included-content  OCTET STRING,
  referenced-content    ReferencedContent
}

ReferencedContent ::= SEQUENCE
{
  content-reference     ContentReference,
  content-size  [60] INTEGER OPTIONAL,
  content-cache-priority    [61] INTEGER (0..255) DEFAULT 127
}

-- . Link Class ____________________________________________________

LinkClass ::= SET
{
  COMPONENTS OF IngredientClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT}),
  link-condition    [62] LinkCondition,
  link-effect   [63] ActionClass
}

LinkCondition ::= SEQUENCE
{
  event-source  ObjectReference,
  event-type    EventType,
  event-data    EventData OPTIONAL
}

EventType ::= ENUMERATED
{
  is-available(1),
  content-available(2),
  is-deleted(3),
  is-running(4),
  is-stopped(5),
  user-input(6),
  anchor-fired(7),
  timer-fired(8),
  asynch-stopped(9),
  interaction-completed(10),
  token-moved-from(11),
  token-moved-to(12),
  stream-event(13),
  stream-playing(14),
  stream-stopped(15),
  counter-trigger(16),
  highlight-on(17),
  highlight-off(18),
  cursor-enter(19),
  cursor-leave(20),
  is-selected(21),
  is-deselected(22),
  test-event(23),
  first-item-presented(24),
  last-item-presented(25),
  head-items(26),
  tail-items(27),
  item-selected(28),
  item-deselected(29),
  entry-field-full(30),
  engine-event(31)
}

EventData ::= CHOICE
{
  octetstring   OCTET STRING,
  boolean   BOOLEAN,
  integer   INTEGER
}

--  Program Class ___________________________________________________
```

```
ProgramClass ::= SET
{
  COMPONENTS OF IngredientClass
    (WITH COMPONENTS {..., initially-active (FALSE) PRESENT}),
  name  [64] OCTET STRING,
  initially-available   [65] BOOLEAN DEFAULT TRUE
}

--  Resident Program Class _____________________________________________

ResidentProgramClass ::= ProgramClass
  (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT})

--  Remote Program Class _____________________________________________

RemoteProgramClass ::= SET
{
  COMPONENTS OF ProgramClass
  (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT}),
  program-connection-tag    [66] INTEGER OPTIONAL
}

--  Interchanged Program Class _____________________________________

InterchangedProgramClass ::= ProgramClass
  (WITH COMPONENTS {..., original-content PRESENT})

--  Palette Class _____________________________________________

PaletteClass ::= IngredientClass
  (WITH COMPONENTS
    {..., content-hook PRESENT, original-content PRESENT, initially-active (TRUE)})

--  Font Class _____________________________________________

FontClass ::= IngredientClass
  (WITH COMPONENTS
    {..., content-hook PRESENT, original-content PRESENT, initially-active (TRUE)})

--  Cursor Shape _____________________________________________

CursorShapeClass ::= IngredientClass
  (WITH COMPONENTS
    {..., content-hook PRESENT, original-content PRESENT, initially-active (TRUE)})

--  Variable Class _____________________________________________

VariableClass ::= SET
{
  COMPONENTS OF IngredientClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT, initially-active
(TRUE)}),
  original-value    [67] OriginalValue
}

OriginalValue ::= CHOICE
{
  boolean   BOOLEAN,
  integer   INTEGER,
  octetstring   OCTET STRING,
  object-reference  [68] ObjectReference,
  content-reference     [69] ContentReference
}

--  Boolean Variable Class _____________________________________

BooleanVariableClass ::= VariableClass
  (WITH COMPONENTS {..., original-value (WITH COMPONENTS {..., boolean PRESENT})})

--  Integer Variable Class _____________________________________

IntegerVariableClass ::= VariableClass
  (WITH COMPONENTS {..., original-value (WITH COMPONENTS {..., integer PRESENT})})

--  Octet String Variable Class _____________________________________

OctetStringVariableClass ::= VariableClass
  (WITH COMPONENTS {..., original-value (WITH COMPONENTS {..., octetstring PRESENT})})

--  Object Reference Variable Class _____________________________________

ObjectRefVariableClass ::= VariableClass
  (WITH COMPONENTS {..., original-value (WITH COMPONENTS {..., object-reference PRESENT})})

--  Content Reference Variable Class _____________________________________
```

```
ContentRefVariableClass ::= VariableClass
  (WITH COMPONENTS {..., original-value (WITH COMPONENTS {..., content-reference PRESENT})})

--  Presentable Class _____________________________________________

PresentableClass ::= IngredientClass

--  Token Manager Class ____________________________________________

TokenManagerClass ::= SET
{
  movement-table    [70] SEQUENCE SIZE (1..MAX) OF Movement OPTIONAL
}

Movement ::= SEQUENCE SIZE (1..MAX) OF INTEGER

--  Token Group Class ______________________________________________

TokenGroupClass ::= SET
{
  COMPONENTS OF PresentableClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT}),
  COMPONENTS OF TokenManagerClass,
  token-group-items     [71] SEQUENCE SIZE (1..MAX) OF TokenGroupItem,
  no-token-action-slots     [72] SEQUENCE SIZE (1..MAX) OF ActionSlot OPTIONAL
}

TokenGroupItem ::= SEQUENCE
{
  a-visible ObjectReference,
  action-slots  SEQUENCE SIZE (1..MAX) OF ActionSlot OPTIONAL
}

ActionSlot ::= CHOICE
{
  action-class  ActionClass,
  null  NULL
}

--  List Group Class _______________________________________________

ListGroupClass ::= SET
{
  COMPONENTS OF TokenGroupClass,
  positions     [73] SEQUENCE SIZE (1..MAX) OF XYPosition,
  wrap-around   [74] BOOLEAN DEFAULT FALSE,
  multiple-selection    [75] BOOLEAN DEFAULT FALSE
}

--  Visible Class __________________________________________________

VisibleClass ::= SET
{
  COMPONENTS OF PresentableClass,
  original-box-size     [76] OriginalBoxSize,
  original-position     [77] XYPosition DEFAULT {x-position 0, y-position 0},
  original-palette-ref  [78] ObjectReference OPTIONAL
}

OriginalBoxSize ::= SEQUENCE
{
  x-length  INTEGER (0..MAX),
  y-length  INTEGER (0..MAX)
}

--  Bitmap Class ___________________________________________________

BitmapClass ::= SET
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., original-content PRESENT}),
  tiling    [79] BOOLEAN DEFAULT FALSE,
  original-transparency     [80] INTEGER (0..100) DEFAULT 0
}

--  Line Art Class _________________________________________________

LineArtClass ::= SET
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., original-content PRESENT}),
  bordered-bounding-box     [81] BOOLEAN DEFAULT TRUE,
  original-line-width   [82] INTEGER DEFAULT 1,
  original-line-style   [83] INTEGER {solid(1), dashed(2), dotted(3)} DEFAULT solid,
```

```
  original-ref-line-colour  [84] Colour OPTIONAL,
  original-ref-fill-colour  [85] Colour OPTIONAL
}

--  Rectangle Class _____________________________________________

RectangleClass ::= LineArtClass
  (WITH COMPONENTS
    {..., content-hook ABSENT, original-content ABSENT, bordered-bounding-box ABSENT})

--  Dynamic Line Art Class ______________________________________

DynamicLineArtClass ::= LineArtClass
  (WITH COMPONENTS
    {..., content-hook ABSENT, original-content ABSENT})

--  Text Class __________________________________________________

TextClass ::= SET
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., original-content PRESENT}),
  original-font     [86] FontBody OPTIONAL,
  font-attributes   [43] OCTET STRING OPTIONAL,
  text-colour   [41] Colour OPTIONAL,
  background-colour     [39] Colour OPTIONAL,
  character-set     [38] INTEGER OPTIONAL,
  horizontal-justification  [87] Justification DEFAULT start,
  vertical-justification    [88] Justification DEFAULT start,
  line-orientation  [89] LineOrientation DEFAULT horizontal,
  start-corner  [90] StartCorner DEFAULT upper-left,
  text-wrapping     [91] BOOLEAN DEFAULT FALSE
}

Justification ::= ENUMERATED
{
  start(1),
  end(2),
  centre(3),
  justified(4)
}

LineOrientation ::= ENUMERATED {vertical(1), horizontal(2)}

StartCorner ::= ENUMERATED
{
  upper-left(1),
  upper-right(2),
  lower-left(3),
  lower-right(4)
}

--  Stream Class ________________________________________________

StreamClass ::= SET
{
  COMPONENTS OF PresentableClass
    (WITH COMPONENTS {..., original-content PRESENT}),
  multiplex     [92] SEQUENCE SIZE (1..MAX) OF StreamComponent,
  storage   [93] Storage DEFAULT stream,
  looping   [94] INTEGER {infinity(0)} DEFAULT 1
}

StreamComponent ::= CHOICE
{
  audio     [95] AudioClass,
  video     [96] VideoClass,
  rtgraphics    [97] RTGraphicsClass
}

Storage ::= ENUMERATED {memory(1), stream(2)}

--  Audio Class _________________________________________________

AudioClass ::= SET
{
  COMPONENTS OF PresentableClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT, shared ABSENT}),
  component-tag     [98] INTEGER,
  original-volume   [99] INTEGER DEFAULT 0
}

--  Video Class _________________________________________________

VideoClass ::= SET
```

```
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT, shared ABSENT, original-
palette-ref ABSENT}),
  component-tag     [98] INTEGER,
  termination   [100] Termination DEFAULT disappear
}

Termination ::= ENUMERATED {freeze(1), disappear(2)}

--  RTGraphics Class ____________________________________________

RTGraphicsClass ::= SET
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT, shared ABSENT}),
  component-tag     [98] INTEGER,
  termination   [100] Termination DEFAULT disappear
}

--  Interactible Class __________________________________________

InteractibleClass ::= SET
{
  engine-resp   [101] BOOLEAN DEFAULT TRUE,
  highlight-ref-colour  [49] Colour OPTIONAL
}

--  Slider Class ________________________________________________

SliderClass ::= SET
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT}),
  COMPONENTS OF InteractibleClass,
  orientation   [102] Orientation,
  max-value     [103] INTEGER,
  min-value     [104] INTEGER DEFAULT 1,
  initial-value     [105] INTEGER OPTIONAL,
  initial-portion   [106] INTEGER OPTIONAL,
  step-size     [107] INTEGER DEFAULT 1,
  slider-style  [108] SliderStyle DEFAULT normal,
  slider-ref-colour     [50] Colour OPTIONAL
}

Orientation ::= ENUMERATED {left(1), right(2), up(3), down(4)}

SliderStyle ::= ENUMERATED {normal(1), thermometer(2), proportional(3)}

--  Entry Field Class ___________________________________________

EntryFieldClass ::= SET
{
  COMPONENTS OF TextClass,
  COMPONENTS OF InteractibleClass,
  input-type    [109] InputType DEFAULT any,
  char-list     [110] OCTET STRING OPTIONAL,
  obscured-input    [111] BOOLEAN DEFAULT FALSE,
  max-length    [112] INTEGER DEFAULT 0
}

InputType ::= ENUMERATED {alpha(1), numeric(2), any(3), listed(4)}

--  Hyper Text Class ____________________________________________

HyperTextClass ::= SET
{
  COMPONENTS OF TextClass,
  COMPONENTS OF InteractibleClass
}

--  Button Class ________________________________________________

ButtonClass ::= SET
{
  COMPONENTS OF VisibleClass
    (WITH COMPONENTS {..., content-hook ABSENT, original-content ABSENT}),
  COMPONENTS OF InteractibleClass,
  button-ref-colour     [48] Colour OPTIONAL
}

--  Hotspot Class _______________________________________________

HotspotClass ::= ButtonClass
```

```
--  Push Button Class _____________________________________________

PushButtonClass ::= SET
{
  COMPONENTS OF ButtonClass,
  original-label    [113] OCTET STRING OPTIONAL,
  character-set     [38] INTEGER OPTIONAL
}

--  Switch Button Class ____________________________________________

SwitchButtonClass ::= SET
{
  COMPONENTS OF PushButtonClass,
  button-style  [114] ButtonStyle
}

ButtonStyle ::= ENUMERATED
{
  pushbutton(1),
  radiobutton(2),
  checkbox(3)
}

--  Action Class ___________________________________________________

ActionClass ::= SEQUENCE SIZE (1..MAX) OF ElementaryAction

ElementaryAction ::= CHOICE
{
  activate  [115] GenericObjectReference,
  add   [116] Add,
  add-item  [117] AddItem,
  append    [118] Append,
  bring-to-front    [119] GenericObjectReference,
  call  [120] Call,
  call-action-slot  [121] CallActionSlot,
  clear     [122] GenericObjectReference,
  clone     [123] Clone,
  close-connection  [124] CloseConnection,
  deactivate    [125] GenericObjectReference,
  del-item  [126] DelItem,
  deselect  [127] GenericObjectReference,
  deselect-item     [128] DeselectItem,
  divide    [129] Divide,
  draw-arc  [130] DrawArc,
  draw-line     [131] DrawLine,
  draw-oval     [132] DrawOval,
  draw-polygon  [133] DrawPolygon,
  draw-polyline     [134] DrawPolyline,
  draw-rectangle    [135] DrawRectangle,
  draw-sector   [136] DrawSector,
  fork  [137] Fork,
  get-availability-status   [138] GetAvailabilityStatus,
  get-box-size  [139] GetBoxSize,
  get-cell-item     [140] GetCellItem,
  get-cursor-position   [141] GetCursorPosition,
  get-engine-support    [142] GetEngineSupport,
  get-entry-point   [143] GetEntryPoint,
  get-fill-colour   [144] GetFillColour,
  get-first-item    [145] GetFirstItem,
  get-highlight-status  [146] GetHighlightStatus,
  get-interaction-status    [147] GetInteractionStatus,
  get-item-status   [148] GetItemStatus,
  get-label     [149] GetLabel,
  get-last-anchor-fired     [150] GetLastAnchorFired,
  get-line-colour   [151] GetLineColour,
  get-line-style    [152] GetLineStyle,
  get-line-width    [153] GetLineWidth,
  get-list-item     [154] GetListItem,
  get-list-size     [155] GetListSize,
  get-overwrite-mode    [156] GetOverwriteMode,
  get-portion   [157] GetPortion,
  get-position  [158] GetPosition,
  get-running-status    [159] GetRunningStatus,
  get-selection-status  [160] GetSelectionStatus,
  get-slider-value  [161] GetSliderValue,
  get-text-content  [162] GetTextContent,
  get-text-data     [163] GetTextData,
  get-token-position    [164] GetTokenPosition,
  get-volume    [165] GetVolume,
  launch    [166] GenericObjectReference,
  lock-screen   [167] GenericObjectReference,
  modulo    [168] Modulo,
  move  [169] Move,
```

```
  move-to   [170] MoveTo,
  multiply  [171] Multiply,
  open-connection   [172] OpenConnection,
  preload   [173] GenericObjectReference,
  put-before    [174] PutBefore,
  put-behind    [175] PutBehind,
  quit  [176] GenericObjectReference,
  read-persistent   [177] ReadPersistent,
  run   [178] GenericObjectReference,
  scale-bitmap  [179] ScaleBitmap,
  scale-video   [180] ScaleVideo,
  scroll-items  [181] ScrollItems,
  select    [182] GenericObjectReference,
  select-item   [183] SelectItem,
  send-event    [184] SendEvent,
  send-to-back  [185] GenericObjectReference,
  set-box-size  [186] SetBoxSize,
  set-cache-priority    [187] SetCachePriority,
  set-counter-end-position  [188] SetCounterEndPosition,
  set-counter-position  [189] SetCounterPosition,
  set-counter-trigger   [190] SetCounterTrigger,
  set-cursor-position   [191] SetCursorPosition,
  set-cursor-shape  [192] SetCursorShape,
  set-data  [193] SetData,
  set-entry-point   [194] SetEntryPoint,
  set-fill-colour   [195] SetFillColour,
  set-first-item    [196] SetFirstItem,
  set-font-ref  [197] SetFontRef,
  set-highlight-status  [198] SetHighlightStatus,
  set-interaction-status    [199] SetInteractionStatus,
  set-label     [200] SetLabel,
  set-line-colour   [201] SetLineColour,
  set-line-style    [202] SetLineStyle,
  set-line-width    [203] SetLineWidth,
  set-overwrite-mode    [204] SetOverwriteMode,
  set-palette-ref   [205] SetPaletteRef,
  set-portion   [206] SetPortion,
  set-position  [207] SetPosition,
  set-slider-value  [208] SetSliderValue,
  set-speed     [209] SetSpeed,
  set-timer     [210] SetTimer,
  set-transparency  [211] SetTransparency,
  set-variable  [212] SetVariable,
  set-volume    [213] SetVolume,
  spawn     [214] GenericObjectReference,
  step  [215] Step,
  stop  [216] GenericObjectReference,
  store-persistent  [217] StorePersistent,
  subtract  [218] Subtract,
  test-variable     [219] TestVariable,
  toggle    [220] GenericObjectReference,
  toggle-item   [221] ToggleItem,
  transition-to     [222] TransitionTo,
  unload    [223] GenericObjectReference,
  unlock-screen     [224] GenericObjectReference
}

Add ::= SEQUENCE
{
  target    GenericObjectReference,
  value     GenericInteger
}

AddItem ::= SEQUENCE
{
  target    GenericObjectReference,
  item-index    GenericInteger,
  visible-reference     GenericObjectReference
}

Append ::= SEQUENCE
{
  target    GenericObjectReference,
  append-value  GenericOctetString
}

Call ::= SEQUENCE
{
  target    GenericObjectReference,
  call-succeeded    ObjectReference,
  parameters    SEQUENCE SIZE (1..MAX) OF Parameter OPTIONAL
}

CallActionSlot ::= SEQUENCE
{
```

```
  target    GenericObjectReference,
  index     GenericInteger
}

Clone ::= SEQUENCE
{
  target    GenericObjectReference,
  clone-ref-var     ObjectReference
}

CloseConnection ::= SEQUENCE
{
  target    GenericObjectReference,
  connection-tag    GenericInteger
}

DelItem ::= SEQUENCE
{
  target    GenericObjectReference,
  visible-reference     GenericObjectReference
}

DeselectItem ::= SEQUENCE
{
  target    GenericObjectReference,
  item-index    GenericInteger
}

Divide ::= SEQUENCE
{
  target    GenericObjectReference,
  value     GenericInteger
}

DrawArc ::= SEQUENCE
{
  target    GenericObjectReference,
  x     GenericInteger,
  y     GenericInteger,
  ellipse-width     GenericInteger,
  ellipse-height    GenericInteger,
  start-angle   GenericInteger,
  arc-angle     GenericInteger
}

DrawLine ::= SEQUENCE
{
  target GenericObjectReference,
  x1    GenericInteger,
  y1    GenericInteger,
  x2    GenericInteger,
  y2    GenericInteger
}

DrawOval ::= SEQUENCE
{
  target    GenericObjectReference,
  x     GenericInteger,
  y     GenericInteger,
  ellipse-width     GenericInteger,
  ellipse-height    GenericInteger
}

DrawPolygon ::= SEQUENCE
{
  target    GenericObjectReference,
  pointlist     SEQUENCE SIZE (1..MAX) OF Point
}

DrawPolyline ::= SEQUENCE
{
  target    GenericObjectReference,
  pointlist     SEQUENCE SIZE (1..MAX) OF Point
}

DrawRectangle ::= SEQUENCE
{
  target    GenericObjectReference,
  x1    GenericInteger,
  y1    GenericInteger,
  x2    GenericInteger,
  y2    GenericInteger
}

DrawSector ::= SEQUENCE
```

```
{
  target    GenericObjectReference,
  x     GenericInteger,
  y     GenericInteger,
  ellipse-width     GenericInteger,
  ellipse-height    GenericInteger,
  start-angle   GenericInteger,
  arc-angle     GenericInteger
}

Fork ::= SEQUENCE
{
  target    GenericObjectReference,
  fork-succeeded    ObjectReference,
  parameters    SEQUENCE SIZE (1..MAX) OF Parameter OPTIONAL
}

GetAvailabilityStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  availability-status-var   ObjectReference
}

GetBoxSize ::= SEQUENCE
{
  target    GenericObjectReference,
  x-box-size-var    ObjectReference,
  y-box-size-var    ObjectReference
}

GetCellItem ::= SEQUENCE
{
  target    GenericObjectReference,
  cell-index    GenericInteger,
  item-ref-var  ObjectReference
}

GetCursorPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  x-out     ObjectReference,
  y-out     ObjectReference
}

GetEngineSupport ::= SEQUENCE
{
  target    GenericObjectReference,
  feature   GenericOctetString,
  answer    ObjectReference
}

GetEntryPoint ::= SEQUENCE
{
  target    GenericObjectReference,
  entry-point-var   ObjectReference
}

GetFillColour ::= SEQUENCE
{
  target    GenericObjectReference,
  fill-colour-var   ObjectReference
}

GetFirstItem ::= SEQUENCE
{
  target    GenericObjectReference,
  first-item-var    ObjectReference
}

GetHighlightStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  highlight-status-var  ObjectReference
}

GetInteractionStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  interaction-status-var    ObjectReference
}

GetItemStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  item-index    GenericInteger,
```

```
  item-status-var   ObjectReference
}

GetLabel ::= SEQUENCE
{
  target    GenericObjectReference,
  label-var     ObjectReference
}

GetLastAnchorFired ::= SEQUENCE
{
  target    GenericObjectReference,
  last-anchor-fired-var     ObjectReference
}

GetLineColour ::= SEQUENCE
{
  target    GenericObjectReference,
  line-colour-var   ObjectReference
}

GetLineStyle ::= SEQUENCE
{
  target    GenericObjectReference,
  line-style-var    ObjectReference
}

GetLineWidth ::= SEQUENCE
{
  target    GenericObjectReference,
  line-width-var    ObjectReference
}

GetListItem ::= SEQUENCE
{
  target    GenericObjectReference,
  item-index    GenericInteger,
  item-ref-var  ObjectReference
}

GetListSize ::= SEQUENCE
{
  target    GenericObjectReference,
  size-var  ObjectReference
}

GetOverwriteMode ::= SEQUENCE
{
  target    GenericObjectReference,
  overwrite-mode-var    ObjectReference
}

GetPortion ::= SEQUENCE
{
  target    GenericObjectReference,
  portion-var   ObjectReference
}

GetPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  x-position-var    ObjectReference,
  y-position-var    ObjectReference
}

GetRunningStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  running-status-var    ObjectReference
}

GetSelectionStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  selection-status-var  ObjectReference
}

GetSliderValue ::= SEQUENCE
{
  target    GenericObjectReference,
  slider-value-var  ObjectReference
}

GetTextContent ::= SEQUENCE
{
```

```
  target    GenericObjectReference,
  text-content-var  ObjectReference
}

GetTextData ::= SEQUENCE
{
  target    GenericObjectReference,
  text-data-var     ObjectReference
}

GetTokenPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  token-position-var    ObjectReference
}

GetVolume ::= SEQUENCE
{
  target    GenericObjectReference,
  volume-var    ObjectReference
}

Modulo ::= SEQUENCE
{
  target    GenericObjectReference,
  value     GenericInteger
}

Move ::= SEQUENCE
{
  target    GenericObjectReference,
  movement-identifier   GenericInteger
}

MoveTo ::= SEQUENCE
{
  target    GenericObjectReference,
  index     GenericInteger
}

Multiply ::= SEQUENCE
{
  target    GenericObjectReference,
  value     GenericInteger
}

OpenConnection ::= SEQUENCE
{
  target    GenericObjectReference,
  open-succeeded    ObjectReference,
  protocol  GenericOctetString,
  address   GenericOctetString,
  connection-tag    GenericInteger
}

PutBefore ::= SEQUENCE
{
  target    GenericObjectReference,
  reference-visible     GenericObjectReference
}

PutBehind ::= SEQUENCE
{
  target    GenericObjectReference,
  reference-visible     GenericObjectReference
}

ReadPersistent ::= SEQUENCE
{
  target    GenericObjectReference,
  read-succeeded    ObjectReference,
  out-variables     SEQUENCE SIZE (1..MAX) OF ObjectReference,
  in-file-name  GenericOctetString
}

ScaleBitmap ::= SEQUENCE
{
  target    GenericObjectReference,
  x-scale   GenericInteger,
  y-scale   GenericInteger
}

ScaleVideo ::= SEQUENCE
{
  target    GenericObjectReference,
```

```
  x-scale   GenericInteger,
  y-scale   GenericInteger
}

ScrollItems ::= SEQUENCE
{
  target    GenericObjectReference,
  items-to-scroll   GenericInteger
}

SelectItem ::= SEQUENCE
{
  target    GenericObjectReference,
  item-index    GenericInteger
}

SendEvent ::= SEQUENCE
{
  target    GenericObjectReference,
  emulated-event-source     GenericObjectReference,
  emulated-event-type   EventType,
  emulated-event-data   EmulatedEventData OPTIONAL
}

SetBoxSize ::= SEQUENCE
{
  target    GenericObjectReference,
  x-new-box-size    GenericInteger,
  y-new-box-size    GenericInteger
}

SetCachePriority ::= SEQUENCE
{
  target    GenericObjectReference,
  new-cache-priority    GenericInteger
}

SetCounterEndPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  new-counter-end-position  GenericInteger
}

SetCounterPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  new-counter-position  GenericInteger
}

SetCounterTrigger ::= SEQUENCE
{
  target    GenericObjectReference,
  trigger-identifier    GenericInteger,
  new-counter-value     GenericInteger OPTIONAL
}

SetCursorPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  x-cursor  GenericInteger,
  y-cursor  GenericInteger
}

SetCursorShape ::= SEQUENCE
{
  target    GenericObjectReference,
  new-cursor-shape  GenericObjectReference OPTIONAL
}

SetData ::= SEQUENCE
{
  target    GenericObjectReference,
  new-content   NewContent
}

SetEntryPoint ::= SEQUENCE
{
  target    GenericObjectReference,
  new-entry-point   GenericInteger
}

SetFillColour ::= SEQUENCE
{
  target    GenericObjectReference,
  new-fill-colour   NewColour OPTIONAL
```

```
}

SetFirstItem ::= SEQUENCE
{
  target    GenericObjectReference,
  new-first-item    GenericInteger
}

SetFontRef ::= SEQUENCE
{
  target    GenericObjectReference,
  new-font  NewFont
}

SetHighlightStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  new-highlight-status  GenericBoolean
}

SetInteractionStatus ::= SEQUENCE
{
  target    GenericObjectReference,
  new-interaction-status    GenericBoolean
}

SetLabel ::= SEQUENCE
{
  target    GenericObjectReference,
  new-label     GenericOctetString
}

SetLineColour ::= SEQUENCE
{
  target    GenericObjectReference,
  new-line-colour   NewColour
}

SetLineStyle ::= SEQUENCE
{
  target    GenericObjectReference,
  new-line-style    GenericInteger
}

SetLineWidth ::= SEQUENCE
{
  target    GenericObjectReference,
  new-line-width    GenericInteger
}

SetOverwriteMode ::= SEQUENCE
{
  target    GenericObjectReference,
  new-overwrite-mode    GenericBoolean
}

SetPaletteRef ::= SEQUENCE
{
  target    GenericObjectReference,
  new-palette-ref   GenericObjectReference
}

SetPortion ::= SEQUENCE
{
  target    GenericObjectReference,
  new-portion   GenericInteger
}

SetPosition ::= SEQUENCE
{
  target    GenericObjectReference,
  new-x-position    GenericInteger,
  new-y-position    GenericInteger
}

SetSliderValue ::= SEQUENCE
{
  target    GenericObjectReference,
  new-slider-value  GenericInteger
}

SetSpeed ::= SEQUENCE
{
  target    GenericObjectReference,
  new-speed     Rational
```

```
}

SetTimer ::= SEQUENCE
{
  target    GenericObjectReference,
  timer-id  GenericInteger,
  new-timer     NewTimer OPTIONAL
}

NewTimer ::= SEQUENCE
{
  timer-value   GenericInteger,
  absolute-time     GenericBoolean OPTIONAL
}

SetTransparency ::= SEQUENCE
{
  target    GenericObjectReference,
  new-transparency  GenericInteger
}

SetVariable ::= SEQUENCE
{
  target    GenericObjectReference,
  new-variable-value    NewVariableValue
}

SetVolume ::= SEQUENCE
{
  target    GenericObjectReference,
  new-volume    GenericInteger
}

Step ::= SEQUENCE
{
  target    GenericObjectReference,
  nb-of-steps   GenericInteger
}

StorePersistent ::= SEQUENCE
{
  target    GenericObjectReference,
  store-succeeded   ObjectReference,
  in-variables  SEQUENCE SIZE (1..MAX) OF ObjectReference,
  out-file-name     GenericOctetString
}

Subtract ::= SEQUENCE
{
  target    GenericObjectReference,
  value     GenericInteger
}

TestVariable ::= SEQUENCE
{
  target    GenericObjectReference,
  operator  GenericInteger,
  comparison-value  ComparisonValue
}

ToggleItem ::= SEQUENCE
{
  target    GenericObjectReference,
  item-index    GenericInteger
}

TransitionTo ::= SEQUENCE
{
 target     GenericObjectReference,
 connection-tag-or-null     ConnectionTagOrNull,
 transition-effect      GenericInteger OPTIONAL
}

ConnectionTagOrNull ::= CHOICE
{
 connection-tag     GenericInteger,
 null       NULL
}

ComparisonValue ::= CHOICE
{
  new-generic-boolean   [225] GenericBoolean,
  new-generic-integer   [226] GenericInteger,
  new-generic-octetstring   [227] GenericOctetString,
```

```
  new-generic-object-reference  [228] GenericObjectReference,
  new-generic-content-reference     [229] GenericContentReference
}

EmulatedEventData ::= CHOICE
{
  new-generic-boolean   [225] GenericBoolean,
  new-generic-integer   [226] GenericInteger,
  new-generic-octet-string  [227] GenericOctetString
}

NewColour ::= CHOICE
{
  new-colour-index  [230] GenericInteger,
  new-absolute-colour   [231] GenericOctetString
}

NewContent ::= CHOICE
{
  new-included-content  GenericOctetString,
  new-referenced-content    NewReferencedContent
}

NewFont ::= CHOICE
{
  new-font-name     [232] GenericOctetString,
  new-font-reference    [233] GenericObjectReference
}

NewReferencedContent ::= SEQUENCE
{
  generic-content-reference     GenericContentReference,
  new-content-size  [234] NewContentSize,
  new-content-cache-priority    [235] GenericInteger OPTIONAL
}

NewContentSize ::= CHOICE
{
  content-size  GenericInteger,
  null  NULL
}

NewVariableValue ::= CHOICE
{
  new-generic-integer   [226] GenericInteger,
  new-generic-boolean   [225] GenericBoolean,
  new-generic-octet-string  [227] GenericOctetString,
  new-generic-object-reference  [228] GenericObjectReference,
  new-generic-content-reference     [229] GenericContentReference
}

Parameter ::= CHOICE
{
  new-generic-boolean   [225] GenericBoolean,
  new-generic-integer   [226] GenericInteger,
  new-generic-octetstring   [227] GenericOctetString,
  new-generic-object-reference  [228] GenericObjectReference,
  new-generic-content-reference     [229] GenericContentReference
}

Point ::= SEQUENCE
{
  x     GenericInteger,
  y     GenericInteger
}

Rational ::= SEQUENCE
{
  numerator     GenericInteger,
  denominator   GenericInteger OPTIONAL
}

--  Referencing Objects, Contents, Values, Colour and Position ____

ObjectReference ::= CHOICE
{
  external-reference    ExternalReference,
  internal-reference    INTEGER (1..MAX)
}

ExternalReference ::= SEQUENCE
{
  group-identifier  OCTET STRING,
  object-number     INTEGER (0..MAX)
}
```

```
IndirectReference ::=   [236] ObjectReference

ContentReference ::=    OCTET STRING

GenericObjectReference ::= CHOICE
{
  direct-reference  ObjectReference,
  indirect-reference    IndirectReference
}

GenericContentReference ::= CHOICE
{
  content-reference     [69] ContentReference,
  indirect-reference    IndirectReference
}

GenericInteger ::= CHOICE
{
  integer   INTEGER,
  indirect-reference    IndirectReference
}

GenericBoolean ::= CHOICE
{
  boolean   BOOLEAN,
  indirect-reference    IndirectReference
}

GenericOctetString ::= CHOICE
{
  octetstring   OCTET STRING,
  indirect-reference    IndirectReference
}

Colour ::= CHOICE
{
  colour-index  INTEGER,
  absolute-colour   OCTET STRING
}

XYPosition ::= SEQUENCE
{
  x-position    INTEGER,
  y-position    INTEGER
}

END
```

History

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